








Modelling Financial Markets

Chapter 15: LLMs and Sentiment Analysis



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Opening: Can We Trade the Sentiment That Models Read in the News?

> *"I find that high media pessimism predicts downward pressure on market prices followed by a reversion to fundamentals, and unusually high or low pessimism predicts high market trading volume."*
>
> — *Tetlock (2007)*

- The chapter compares dictionaries, FinBERT, zero-shot LLMs (large language models) and embeddings on the same labelled texts
  - on Twitter, the best is the simplest supervised model, TF-IDF (weighted word frequencies) + logistic regression: 83.2% accuracy, against 74.2% for Qwen2.5-7B and 72.5% for FinBERT
- On 16 large US stocks with 106 820 news headlines, the score explains the same-day price move
  - FinBERT: +39.2 bp per standard deviation of the score ($t = +17.27$); bp: basis points, 1 bp = 0.01%
- A daily long-short strategy on the sign of the score brings no net profit
  - Qwen2.5-7B: +5.64 bp per day before costs, $p = 0.071$ after the Holm correction
  - the break-even cost, 1.5 bp per trade, is below the assumed cost of 5 bp
- LLMs remember market history
  - asked only for the month, Qwen2.5-14B separates up from down months of the S&P 500 with an AUC of 0.69 (area under the ROC curve; 0.5 = no discrimination)

▸ Central chart: Cumulative Strategy Returns: No Net Profit

Can Models Read News Sentiment, and Can We Trade It?

From Text to a Trading Signal

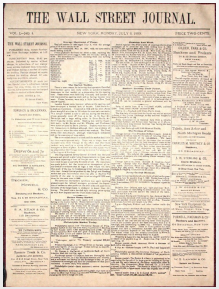
- Part I: text as data, dictionaries, evaluation of classifiers
  - FinBERT and LLMs (large language models), embeddings
- Part II: look-ahead bias, the trading signal, scores as estimates
  - Two case studies and one pass of the research loop assisted by AI (artificial intelligence)
- Detailed examples and advanced topics are in Further Topics, after the Conclusions

What you should be able to do

- compute the sentiment of a text with a dictionary, FinBERT and a zero-shot LLM
- interpret the paired test that compares two classifiers on the same texts
- interpret the tests for training overlap and memorised outcomes in an LLM backtest
- interpret the test of a sentiment signal with correct timing, few-cluster inference, costs and a multiplicity correction
- derive the attenuation of the slope when a model score is a generated regressor
- interpret prediction-powered inference intervals
- Further Topics and Appendix: reading for familiarity

Markets Run on Words

- Prices move when information arrives, and most information arrives as text
  - News, company filings, earnings calls, analyst reports, central-bank statements, social media
- Financial newspapers are as old as organised markets
  - The Wall Street Journal: first issue on 8 July 1889
- Today: millions of documents per day, far more than any analyst can read


THE WALL STREET JOURNAL.

NEW YORK, MONDAY, JULY 8 1889. PRICE TWO CENTS.

The Wall Street Journal, first issue, 8 July 1889

Photo: Dow Jones & Company (1889); public domain; Wikimedia Commons

Text as Data, and What We Mean by Sentiment

- Gentzkow, Kelly & Taddy (2019): a document d becomes numbers x_d (counts, scores, vectors)
  - These numbers feed an ordinary statistical model
  - Map x_d to a quantity of interest (sentiment, topic, risk), then use it in an economic question (returns, volatility)
- **Sentiment of a text**: good, bad or neutral news for the asset it mentions
  - Three classes, or the net score $s = P(\text{positive}) - P(\text{negative}) \in [-1, 1]$
  - Not **investor sentiment**, the mood of investors measured from market data (Baker & Wurgler (2006))
- Surveys for finance: Kearney & Liu (2014), Loughran & McDonald (2016)
- This chapter: the score is an **estimate** of the true tone, with measurement error

Early Evidence and Narratives

- Antweiler & Frank (2004): internet message-board posts about stocks help predict volatility
  - The effect on returns is small
- Tetlock (2007): high media pessimism predicts downward pressure on prices
  - Pessimism measured in a Wall Street Journal column
  - The pressure is followed by a reversal
  - Unusually high or low pessimism also predicts high trading volume
- Tetlock, Saar-Tsechansky & Macskassy (2008): negative words in firm-specific news predict lower earnings and returns
  - Measure: the share of negative words
  - Prices incorporate this information only gradually
- Shiller (2017): popular stories spread like epidemics and move economic decisions
  - They can be measured through text
  - Shiller: Nobel Prize in Economics 2013, with Fama and Hansen; narratives can push prices away from fundamentals (Chapters 2 and 17)



Robert J. Shiller, Stockholm, December 2013

Photo: Bengt Nyman (2013); CC BY 2.0; Wikimedia Commons

Four Ways to Turn Text into Sentiment

Method	How it works	Labels needed	Examples
Dictionary	count listed words	none	GI, LM, VADER
Fine-tuned model	network trained on labelled sentences	thousands	FinBERT
Large language model	instruction in plain language (prompt)	none (zero-shot)	Qwen2.5, GPT
Embeddings + classifier	text vectors into a regression	hundreds to thousands	MiniLM

GI: the Harvard General Inquirer, a general-purpose word list; LM: the Loughran–McDonald financial dictionary; VADER: Valence Aware Dictionary and sEntiment Reasoner, a dictionary with rules built for social media; GPT: Generative Pre-trained Transformer, the language models from OpenAI

- We compare all four on the same labelled data, with 95% confidence intervals (CI)
- Then we test whether any of them makes money on daily stock returns
- A fifth route, text regressions with statistical theory, is not tested here (Appendix)
  - Taddy; Ke, Kelly and Xiu

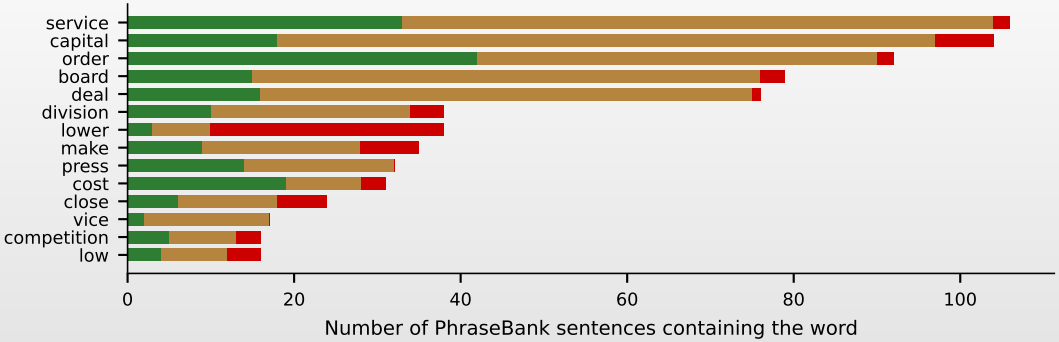
Can a Word List Measure the Tone of a Financial Text?

- **Bag of words**: a document is reduced to its word counts
  - Word order is ignored
  - Tokens: case-insensitive words, without digits and punctuation
- A dictionary lists positive words $\mathcal{P}$ and negative words $\mathcal{N}$
  - P_d, N_d: number of positive and negative words in document d
  - **Tone**: $\tau_d = \dfrac{P_d - N_d}{P_d + N_d} \in [-1, 1]$, and $\tau_d = 0$ if $P_d + N_d = 0$; the class is the sign of τ_d (neutral for $\tau_d = 0$)
- Transparent, fast and free of training data
  - Blind to context and negation

Generic, Financial and Rule-Based Dictionaries

- **Harvard General Inquirer** (GI), word list Harvard IV-4: a psychology dictionary from the 1960s
  - 2 007 negative and 1 637 positive words
- Loughran & McDonald (2011) (LM): built from annual reports (10-K filings) of US firms
  - 2 355 negative and only 354 positive words
  - Their finding: almost three quarters of the GI negative words are not negative in a financial context
  - In the lists used here, 1 552 GI negative words (77%) are not LM negative words
- **VADER** (Hutto & Gilbert (2014)): about 7,500 terms with intensity scores, built for social media
  - Rules for negation ("not good"), intensifiers ("very"), capital letters, exclamation marks and emoticons
  - Compound score in $[-1, 1]$, usual cut-offs ± 0.05
  - The rules fix some context problems, but the vocabulary is generic, like GI

Negative for a Psychologist, Neutral for an Accountant (1/2)



MFM_ch15_dictionaries ▶ Interactive

Negative for a Psychologist, Neutral for an Accountant (2/2)

- Words negative in GI but not in LM
  - Ranked by the number of Financial PhraseBank sentences that contain them
  - Colour: the label human annotators gave the sentence
- "Service" (106 sentences), "capital" (104) and "board" (79) are business nouns, not bad news
  - Not labelled negative: 98%, 93%, 96% of their sentences; "vice" ("vice president"): 100%
- "Lower": only 26% not negative, but "lower costs" is good news
  - The sign depends on the next word, which no word list sees
- Of the 964 sentences whose only GI-negative words are missing from the LM negative list:
  - 89% are not labelled negative

MFM_ch15_dictionaries

Which Texts Do We Judge a Classifier On? Two Labelled Corpora

- Financial PhraseBank (Malo et al. (2014)): 4 846 news sentences
  - Labelled by a pool of 16 annotators with a finance background, 5 to 8 per sentence
  - Question: good, bad or neutral news for the investor?
  - Agreement levels: at least 50%, 66%, 75% or 100% of the sentence's annotators; 2 264 sentences with full agreement, 629 below 66%
  - Labels: 59% neutral, 28% positive, 12% negative
- Twitter Financial News (Hugging Face Hub; MIT licence): headlines from 2022
  - Labelled bearish, bullish or neutral
  - Training set of 9,543, validation set of 2 388 headlines (used for evaluation)
  - Labels: 66% neutral, 20% positive, 15% negative
  - Short, noisy texts with tickers and links: closer to the real news flow
- Two corpora, because a result on one may not carry over to the other

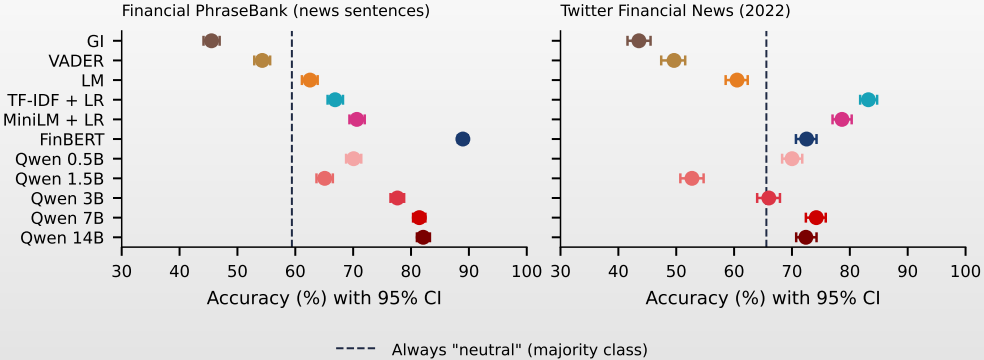
Accuracy or Macro-F1? Metrics for Unbalanced Classes

- The question: how do we compare classifiers when one class dominates?
- Accuracy: $\text{acc} = \frac{1}{n} \sum_{i=1}^{n} 1\{\hat{y}_i = y_i\}$
- For class k: precision $P_k = \frac{TP_k}{TP_k + FP_k}$, recall $R_k = \frac{TP_k}{TP_k + FN_k}$, $F1_k = \frac{2P_kR_k}{P_k + R_k}$
  - TP_k: texts of class k predicted k; FP_k: texts of other classes predicted k; FN_k: texts of class k predicted otherwise
- Macro-F1 $= \frac{1}{K} \sum_{k=1}^{K} F1_k$, with $K = 3$
  - Every class counts the same, however rare (Chapter 13)
- Benchmark "always neutral": 66% accuracy, but a macro-F1 of only 26.4%
  - $F1_{\text{neutral}} = 2 \times 0.656 \times 1/(0.656 + 1) = 0.792$, and $F1 = 0$ for the other two classes, so macro-F1 $= 0.792/3$

When Is One Classifier Better? The McNemar Test on the Same Texts

- Two classifiers A and B on the same n texts: their accuracies are correlated
  - So two separate CIs do not answer the question
- n_{01}: number of texts where A is wrong and B is right
- n_{10}: number of texts where A is right and B is wrong
- Texts on which both classifiers agree carry no information
  - The accuracy gap is $(n_{01} - n_{10})/n$
- Under H_0 (equal accuracy), $n_{01} \sim \text{Binomial}(n_{01} + n_{10}, 0.5)$
  - An exact test (worked example: Appendix)
- For large $n_{01} + n_{10}$: $\chi^2 = (|n_{01} - n_{10}| - 1)^2/(n_{01} + n_{10})$ is approximately χ^2_1
  - -1: continuity correction
- Example: A is wrong and B right on 4 texts, the reverse on 1
  - So $p = 2\,P(X \geq 4) = 2 \times 6/32 = 0.375$, with $X \sim \text{Binomial}(5, 0.5)$

Eleven Classifiers, Two Data Sets: No Single Winner (1/2)



TF-IDF: term frequency–inverse document frequency, word counts weighted down for common words; LR: logistic regression

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Eleven Classifiers, Two Data Sets: No Single Winner (2/2)

- TF-IDF + LR and MiniLM + LR: the only methods trained on Twitter labels
- All three dictionaries lose to the majority class on Twitter
  - Twitter: GI 43.6%, VADER 49.6%, LM 60.5%; "always neutral": 66%
- FinBERT: 89.0% on PhraseBank, 72.5% on Twitter; Qwen2.5-7B zero-shot: 81.4% and 74.2%
- Best on Twitter, 83.2%: the simplest supervised model, TF-IDF + logistic regression
- Lesson: labelled data from the target domain beat model size
- What the result supports: comparison of procedures

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Further Topics: disagreement and kappa

Does Context Help? From Words to BERT

- A dictionary gives "lower" one fixed sign
  - A language model reads the sentence around it
  - Each token gets a vector that depends on all other tokens (attention, Chapter 14; Vaswani et al. (2017))
- **BERT** (Bidirectional Encoder Representations from Transformers; Devlin et al. (2019))
  - A Transformer encoder pretrained on books and Wikipedia by guessing words hidden at random
  - About 110 million parameters
  - For classification it uses the final vector $h_{\mathbf{CLS}} \in \mathbb{R}^{\mathbf{768}}$ of the special token [CLS], placed at the start of every text
- Fine-tuning: a classification layer on top of $h_{\mathbf{CLS}}$
  - The whole network is trained on labelled sentences
  - $p = \text{softmax}(Wh_{\mathbf{CLS}} + c)$, with $W \in \mathbb{R}^{\mathbf{3}\times\mathbf{768}}$: probabilities of the negative, neutral and positive classes
  - Training minimises the cross-entropy $-\frac{\mathbf{1}}{n}\sum_i \ln p_{i,y_i}$, with y_i the human label of sentence i

FinBERT: BERT Adapted to Financial Text

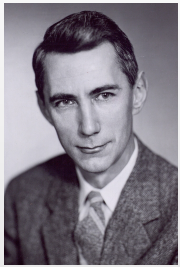
- Araci (2019): FinBERT is BERT further pretrained on financial news, then fine-tuned for sentiment
  - The public model used here (ProsusAI/finbert) was fine-tuned on Financial PhraseBank
  - A few milliseconds per sentence on a laptop
- The score used in this chapter: $s = p_{\text{pos}} - p_{\text{neg}} \in [-1, 1]$
  - The label is $\arg\max_k p_k$
- Why FinBERT as the benchmark: public, fast and widely used in the finance literature
- Huang, Wang & Yang (2023): a FinBERT trained on analyst reports, earnings-call transcripts and 10-K filings
  - It outperforms dictionaries, mainly on sentences that dictionaries misread

The In-Sample Trap

- FinBERT: 89.0% on PhraseBank, most of which it was trained on
  - Araci (2019): 3,101 of 4,845 sentences
  - 72.5% on Twitter: a drop of 16.4 points
  - Qwen2.5-7B, not fine-tuned on these labels, also drops by 7.2 points: Twitter headlines are harder
- Difference in differences: $\Delta = (a_{F,P} - a_{F,T}) - (a_{Q,P} - a_{Q,T}) = 9.2$ points
  - Bootstrap CI 95% [6.7; 11.9]
  - $a_{m,D}$: accuracy of model m (F: FinBERT, Q: Qwen) on data D (P: PhraseBank, T: Twitter)
  - Identifying assumption: without training overlap, both models would lose the same amount from PhraseBank to Twitter
  - The CI measures sampling noise, not the assumption (the two architectures may react differently to a new domain)
- What the result supports: a descriptive difference between the accuracy drops
  - Attributing Δ to training overlap also needs an uncontaminated comparator and no other differential change
  - Qwen was not fine-tuned on PhraseBank, but the texts may be in its pretraining data
- Rule: evaluate a language model only on texts it has never seen
  - As in the out-of-sample tests of Chapters 8 and 13

From Next-Word Prediction to Instructions

- Shannon (1951): English is highly predictable from its past letters
  - A large language model (LLM) estimates $P(w_{t+1} \mid w_1, \ldots, w_t)$ over tokens
  - w_t: the t-th token, a word or part of a word
  - Brown et al. (2020): at scale, tasks are solved from a few examples in the prompt (few-shot)
  - Ouyang et al. (2022): instruction tuning and RLHF (reinforcement learning from human feedback) teach the model to follow requests
- BloombergGPT (Wu et al. (2023)): 50 billion parameters, financial and general text
  - Background: Quantinar: The Next Word Problem, Quantinar: Natural Language Processing



Claude E. Shannon, founder of information theory

Photo: Tekniska museet, Stockholm (undated); CC BY 2.0; Wikimedia Commons

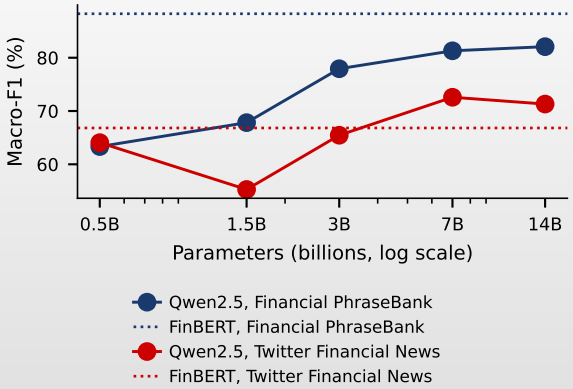
Zero-Shot Classification with Qwen2.5: the Prompt and the Probabilities

- Prompt: a system instruction plus the text to classify
  - `You classify the sentiment of financial news for investors. Answer with one word: positive, negative or neutral.`
  - `Headline: <text> Sentiment:`
- We read the logits z_k of the tokens "positive", "negative", "neutral" at the first generated position
  - $p_k = e^{z_k} / \sum_j e^{z_j}$, normalised over the three answer words
  - Label: $\arg\max_k p_k$
  - Score: $p_{\mathbf{pos}} - p_{\mathbf{neg}}$, as for FinBERT

Temperature and the Qwen2.5 Models

- When generating, the model draws the next token from $p_k(T) = e^{z_k/T} / \sum_j e^{z_j/T}$
  - With temperature $T > 0$
  - $T \to 0$: always the most likely token (greedy decoding), so the same answer on every run; Lopez-Lira & Tang (2026) use temperature 0
  - $T = 1$: the model's own distribution; $T > 1$ flattens the probabilities, so answers vary more
- No text is generated here: the label $\arg\max_k p_k$ is the $T \to 0$ answer
  - The score keeps the $T = 1$ probabilities, with nothing to parse
- Qwen Team (2024): open-weight models, from 0.5 to 72 billion parameters
  - Licences: Apache 2.0; Qwen Research License for 3B
  - Released on 19 September 2024
- We use 0.5B, 1.5B, 3B, 7B and 14B (B: billion parameters)
  - Processing times: Appendix
- Open weights: results can be reproduced
  - The release date bounds what the model could know

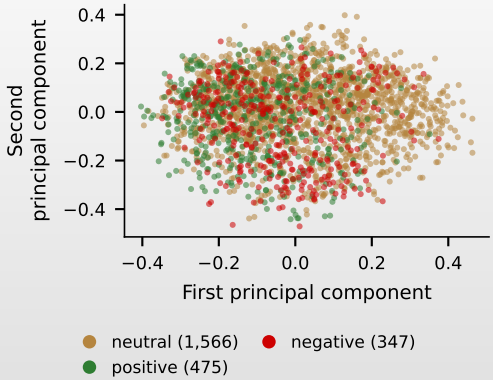
Bigger Is Not Always Better



- Macro-F1 of zero-shot Qwen2.5 with the same prompt; dotted lines: FinBERT
- PhraseBank: macro-F1 rises from 63.3% (0.5B) to 82.1% (14B), still below FinBERT (88.3%)
  - FinBERT's lead there is partly training overlap (The In-Sample Trap)
- Twitter: not monotone
  - 1.5B is the worst (55.3%)
  - 7B the best (72.6%), above FinBERT (66.8%)
  - 7B against 1.5B on accuracy: +21.5 points, McNemar $p < 0.001$
- Small models react strongly to wording
  - The 1.5B model rarely answers "neutral" with this prompt

MFM_ch15_llm_zero_shot ▶ Interactive

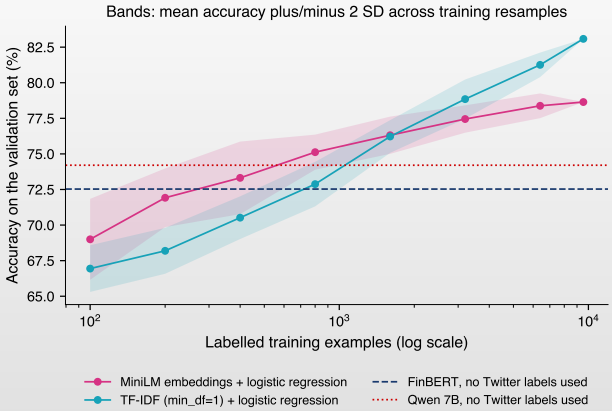
Can Generic Text Vectors Measure Sentiment?



- **Embedding**: a vector $e_d \in \mathbb{R}^k$, close for similar texts
  - word2vec (Mikolov et al. (2013)), Sentence-BERT (Reimers & Gurevych (2019))
  - Here MiniLM (Wang et al. (2020)), $k = 384$, unit length: similarity $\cos(e_a, e_b) = e_a^\top e_b$; embeddings enter a classifier or regression like any feature
- Chart: Twitter validation headlines, first two principal components of the MiniLM vectors
  - Coloured by label
- The first two components do not separate sentiment
  - PCA (principal component analysis, Chapter 3) keeps the directions of largest variance
  - Not those tied to the label
- A logistic regression on all 384 dimensions finds the sentiment direction: 78.6% accuracy on Twitter

MFM_ch15_embeddings ▶ Interactive

How Many Labels Do You Need? (1/2)



MFM_ch15_embeddings ▶ Interactive

How Many Labels Do You Need? (2/2)

- Logistic regression trained on random subsets of the Twitter training set
  - 10 repetitions per size; band: ± 2 standard deviations
- Embeddings + logistic regression: 69.0% with 100 labels, above FinBERT from 400 labels
  - The pretrained vectors already contain most of what is needed
- TF-IDF starts lower (66.9%) but keeps improving
  - About 83% with all labels, against 78.6% for embeddings
  - With many labels, exact words ("upgrade", "cuts", "misses") carry the signal in headlines
- Practical rule: label a few hundred documents from your own domain
  - Do this before trusting any off-the-shelf model

MFM_ch15_embeddings

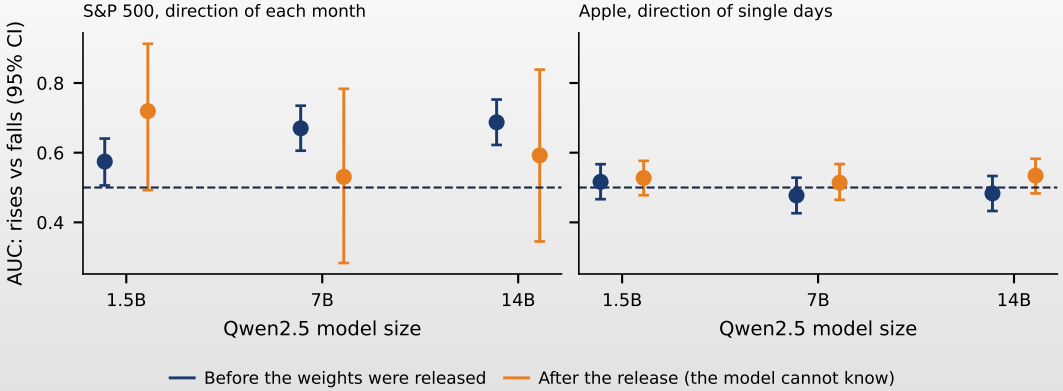
May the Model Already Know the Answer?

- An LLM trained on text up to 2024 may have read what happened after a 2015 headline
  - A backtest on 2015 news then mixes forecasting with remembering
- Glasserman & Lin (2023): two biases in sentiment backtests
  - Look-ahead: the model knows the returns that followed the news
  - Distraction: general knowledge about the named company interferes with reading the text; in their tests it mattered more than look-ahead
- Sarkar & Vafa (2024): direct tests show look-ahead in LLM applications to earnings calls and elections

Our Test: Ask About the Past, Without Context

- Monthly: "Did the S&P 500 index rise or fall during <month year>?"
  - Every month from 296 pre-release months plus 23 after
  - Daily: "Did Apple stock close higher or lower on <date> than on the previous trading day?", 500 random days 2010–2023 and 493 days after October 2024
- Score: $P(\text{rise})$ from the answer-word probabilities
  - True direction from the S&P 500 monthly closes and the Apple daily closes
  - AUC (area under the ROC, receiver operating characteristic, curve): probability that a random up period gets a higher score than a random down period; 0.5 = no discrimination (Chapter 13)
- After the release there is nothing to remember
  - If the date carries no predictive information (e.g. seasonality), the population AUC is 0.5
  - The sample AUC varies around it

Memory Measured: Months Yes, Days No (1/2)



MFM_ch15_lookahead ▶ Interactive

Memory Measured: Months Yes, Days No (2/2)

- AUC with bootstrap 95% intervals, months resampled independently
- Months before release: AUC 0.67 (7B, CI [0.61; 0.73]) and 0.69 (14B, CI [0.62; 0.75])
  - Permutation p for 14B <0.001
  - The 1.5B model knows much less (0.57)
- After release (23 months) the intervals include 0.5 but are too wide to prove anything
  - About 101 post-release months would be needed to detect a fall to 0.5 (formal test: Appendix)
- Single days of Apple: AUC between 0.48 and 0.52 before release, no discrimination above chance detected
  - Look-ahead is a danger at the horizon of well-known events, less for daily noise
- What the result supports: description; the discrimination of months is compatible with memorisation but does not separate it from calendar or regime regularities

MFM_ch15_lookahead

Remedies, Including Point-in-Time Models

- Test only after the model's training cutoff
  - Lopez-Lira & Tang (2026) evaluate GPT-4 on post-cutoff headlines
  - Clean, but the window starts late
- Mask company names and dates before scoring (Glasserman & Lin (2023))
  - Then compare slopes with and without masking
  - A diagnostic, since products, events or quotes can still identify the episode
- Point-in-time models, trained only on text available up to date t (Sarkar & Vafa (2024))
  - Drinkall et al. (2024): a series of point-in-time LLMs, each uninformed about facts after its date; models and training data are public
  - He et al. (2025): ChronoBERT and ChronoGPT use only text available at each date; on next-day returns from news their Sharpe ratios are close to a much larger model: look-ahead there is modest
- Report the model version and the release date of its weights with every result

Dated Headlines Matched with Daily Prices

- The question of this section: does headline tone predict returns that we can still trade?
- FNSPID (Financial News and Stock Price Integration Dataset, Dong, Fan & Peng (2024))
  - 15.7 million news records about S&P 500 firms, 1999–2023, from four news websites
  - Creative Commons licence CC BY-NC 4.0 (attribution, non-commercial use)
- Here: 16 large US stocks with daily prices in the course data
  - Technology, banks, GameStop, Strategy
  - 106 820 distinct headlines after removing repeats; 18 282 stock-days with news, 27.10.2009 to 18.12.2023

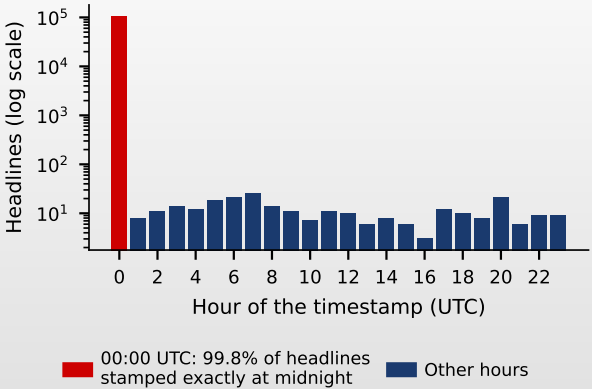


News ticker on the Reuters building, New York

Photo: bgilliard (2005); CC BY-SA 2.0; Wikimedia Commons

▸ Further Topics: headline coverage

The Timestamp Problem: Dates, Not Times (1/2)



MFM_ch15_news_data ▶ Interactive

The Timestamp Problem: Dates, Not Times (2/2)

- Hour of the FNSPID timestamp in Coordinated Universal Time (UTC), log scale
- 99.8% of headlines are stamped exactly 00:00 UTC
  - The source gives the date, not the time of publication
  - A headline dated d may have appeared before the open, during the session or after the close of day d
- Alignment used here, with no look-ahead
  - Day d: the first trading day on or after the date of the headline
  - Returns on d and $d+1$ may still contain the reaction
  - First safe forecast: the return on $d+2$, close of $d+1$ to close of $d+2$

MFM_ch15_news_data

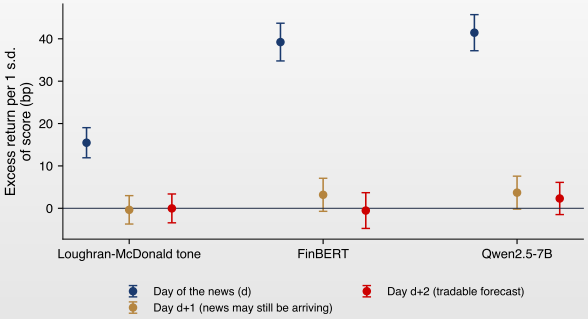
Three Scores and the Excess Return

- The three scores: LM tone, FinBERT net score and Qwen2.5-7B net score with prompt P1
  - FinBERT net score: $P(\text{pos}) - P(\text{neg})$
  - LM finds a listed word in only 27% of headlines: the rest get tone 0
  - FinBERT labels: 19% negative, 62% neutral, 18% positive
  - Correlations across headlines: FinBERT–Qwen 0.62, FinBERT–LM 0.41, Qwen–LM 0.36
- Excess return: $r^{e}_{i,t} = r_{i,t} - r_{\mathbf{SPY},t}$
  - SPY: an exchange-traded fund tracking the S&P 500 index
  - Market-adjusted: $\beta = 1$ for every stock; the usual alternative is $r_{i,t} - \hat{\alpha}_i - \hat{\beta}_i r_{\mathbf{SPY},t}$, with $\hat{\beta}_i$ estimated before the event

The Regression and Standard Errors Clustered by Day

- Pooled regression over stock-days with news: $r^{e}_{i,d+h} = a + b\, z_{i,d} + \varepsilon_{i,d+h}$, $h = 0, 1, 2$
  - z: daily score standardised over the panel (mean 0, standard deviation 1), so b is the excess return (%) per one standard deviation of score
  - $h = 0$ measures reaction, $h = 2$ measures predictability
- Errors of stocks on the same day are correlated (common market shocks)
  - OLS (ordinary least squares) SEs ignore this: cluster by day (Petersen (2009))
  - $\widehat{V} = (X^\top X)^{-1} \left(\sum_g X_g^\top \hat{\varepsilon}_g \hat{\varepsilon}_g^\top X_g \right) (X^\top X)^{-1}$
  - g: trading days; X: the regressors (constant, z); X_g, $\hat{\varepsilon}_g$: the rows and OLS residuals of day g

Reaction versus Prediction: Headlines Describe Prices



- Slope b in basis points (bp, 1 bp = 0.01%) per one standard deviation of score (s.d. in the chart)
  - 95% CI clustered by day
- Day d: FinBERT +39.2 bp ($t = +17.27$), Qwen +41.4 bp ($t = +19.07$), LM +15.5 bp ($t = +8.53$)
  - Many headlines report the price move itself ("shares jump", "stock falls"): causality runs both ways
- Day $d + 1$: FinBERT +3.2 bp ($t = +1.60$), Qwen +3.7 bp ($t = +1.86$)
  - Part of this is news published after the close of d, not a forecast
- Day $d + 2$: FinBERT -0.5 bp ($t = -0.25$), Qwen +2.3 bp ($t = +1.19$), LM -0.0 bp ($t = -0.01$)

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Inference with 16 Stocks: Clustering by Stock and Two-Way

- The question: do the slopes survive if errors are also correlated over time within a stock?
- Firm effects persist, so the errors of one stock are correlated across days: cluster by stock as well
- Two-way clustering (Cameron, Gelbach & Miller (2011)): $\widehat{V}_{\mathbf{2way}} = \widehat{V}_{\mathbf{day}} + \widehat{V}_{\mathbf{stock}} - \widehat{V}_{\mathbf{day} \cap \mathbf{stock}}$
  - Each term is the sandwich formula of the previous slide, with clusters given by day, by stock, and by the stock–day pair
  - A stock–day pair holds a single observation here, so the last term is the White estimator, robust to heteroskedasticity only
  - Subtracting it avoids counting the variance of each observation twice
- Driscoll & Kraay (1998) also allow for correlation across different days

Few Clusters: the Wild Cluster Bootstrap

- With only $G = 16$ stocks, the stock-clustered SE is too small and the t test rejects too often (Cameron, Gelbach & Miller (2008))
- Wild cluster bootstrap with the null imposed: fit the restricted model ($b = 0$)
  - Constant $\tilde{a}$ and residuals $\tilde{\varepsilon}_{i,t}$
  - For $j = 1, \ldots, B$, $B = 9{,}999$: Rademacher weights $w_g = \pm 1$ with probability $\frac{1}{2}$, one draw per stock g
  - $r^{e*}_{i,t} = \tilde{a} + w_{g(i)}\,\tilde{\varepsilon}_{i,t}$; re-fit the regression and compute the stock-clustered t^*_j
  - $p = \frac{1}{B}\sum_j 1\{|t^*_j| \geq |t|\}$
- The same sign for all residuals of a stock keeps the correlation within the cluster
- The bootstrap assumes independent stocks, so read it next to the two-way t

Nine Regressions: the Holm Correction

- Nine regressions: 3 scores $\times$ 3 horizons ($h = 0, 1, 2$)
- If no score had any effect, nine tests at 5% would still give $9 \times 0.05 = 0.45$ false rejections on average
- The Holm correction (Chapter 2) keeps the FWER at or below 5%
  - FWER (family-wise error rate): the chance of any false rejection
  - Sort the p-values: $p_{(1)} \leq \cdots \leq p_{(m)}$, with $m = 9$
  - $\tilde{p}_{(k)} = \max_{j \leq k} \min\{1, (m - j + 1)\, p_{(j)}\}$; reject $H_{(k)}$ when $\tilde{p}_{(k)} \leq 0.05$
  - More powerful than Bonferroni ($m\, p_{(k)}$ for all) and valid under any dependence between the tests

Inference with 16 Stocks and Nine Regressions: Results

Score, horizon	b (bp)	t by stock	t two-way	p wild bootstrap	p Holm
FinBERT, d	+39.2	+5.71	+5.67	<0.001	<0.001
Qwen, d	+41.4	+5.10	+5.07	<0.001	<0.001
FinBERT, $d+2$	−0.5	−0.35	−0.30	0.739	1.00
Qwen, $d+2$	+2.3	+2.41	+2.01	0.061	0.37
LM, $d+2$	−0.0	−0.03	−0.02	0.979	1.00

- Qwen at $d+2$: $t = +2.41$ by stock looks significant
  - $p = 0.029$ under a t distribution with 15 degrees of freedom
  - But not after the wild bootstrap and Holm
  - Holm step by step: sort the 9 bootstrap p-values; Qwen at $d+2$ comes 4th, after the three same-day values (all below 0.001), so its Holm $p = (9 - 4 + 1) \times 0.061 \approx 0.37$
  - Rows not shown: LM at d, p <0.001; at $d+1$, $p = 0.272$ (FinBERT), 0.317 (Qwen), 0.802 (LM)
- What the result supports: association
  - The most positive scores already gain +0.63% in the 5 days before the news (Further Topics)

A Daily Long-Short Sentiment Strategy

- The question: does the sign of the score earn a profit we can actually collect?
- At the close of $d+1$: for every stock with news and a non-zero score on d, go long or short
  - Long if its score is positive and short if negative
  - Each position hedged with SPY; a one-for-one hedge removes the market move only when $\beta = 1$
  - Equal weights; positions are closed after one day, at the close of $d+2$
  - The LM score is zero on 48% of stock-days: these do not enter the portfolio
- Gross return on $d+2$: $G_{d+2} = \frac{1}{N_d} \sum_{i \in A_d} \text{sign}(s_{i,d}) \left(r_{i,d+2} - r_{\text{SPY},d+2}\right)$
  - $A_d = \{i : |s_{i,d}| > b\}$, with threshold $b = 0$: the stocks with news and a non-zero score on day d
  - $N_d = |A_d|$: the positions opened at the close of $d+1$; if A_d is empty, $G_{d+2} = 0$ and no costs are paid

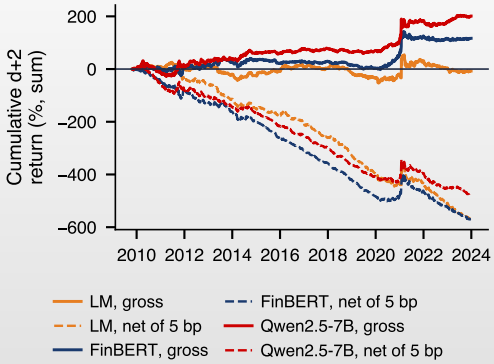
Trading Costs: Round Trips and the Break-Even Cost

- $c = 5$ bp per one-way trade, per unit of traded notional
  - Assumed, not estimated; plausible for large, liquid US stocks
- A round trip (in and out) costs $2c$
- On a day with positions: one round trip for the stock book and one for the SPY hedge
  - Each of notional 1, so $4c = 20$ bp
- Net return: $G^{\text{net}}_{d+2} = G_{d+2} - 4c\,1\{N_d > 0\}$
- Break-even cost c^*: the value of c at which the mean net return is 0
  - I.e. $c^* = \bar{G}_{\text{active}}/4$
  - $\bar{G}_{\text{active}}$: the mean gross return on days with positions

Reading the Strategy Numbers: Sharpe Ratio and Summed Returns

- Inference: Newey–West t (Chapter 2) and block-bootstrap CI for the mean daily return
- Sharpe ratio $SR = \sqrt{252}\,\bar{G}/\hat{\sigma}$ (annualised)
  - $\bar{G}$, $\hat{\sigma}$ are the mean and standard deviation of the daily return G_t over all 3 568 trading days
  - Days without positions enter as 0
  - After costs, G_t is replaced by the net return
  - Example (Qwen2.5-7B): mean $+5.64$ bp and $SR = +0.63$ imply $\hat{\sigma} = \sqrt{252}\times$ mean$/SR \approx 143$ bp per day
  - After costs: positions on 95% of days, so the mean falls by $20\times$ 0.95 bp, to -13.36 bp per day
- The next chart sums daily simple excess returns
  - An additive profit measure, not compounded wealth

Cumulative Strategy Returns: No Net Profit



- Sum of daily excess returns (%), before costs and with 5 bp per trade
- FinBERT: +3.27 bp per day ($t = +1.27$), not different from zero
  - CI 95% [−1.51; +8.93]; LM: −0.21 bp ($t = -0.07$)
- Qwen2.5-7B: +5.64 bp ($t = +2.26$, $p = 0.024$)
  - CI 95% [+0.95; +10.61]; best of three scores: Holm $p = 0.071$, not significant
  - Break-even cost +1.5 bp per trade, well below 5 bp: Sharpe +0.63 before, −1.48 after costs
- What the result supports: association
  - Only 2020–2021 has large gains, not significant at 5% (Further Topics)

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The Strategy: the Non-Tradable Day and Memorisation

- Non-tradable day $d+1$: FinBERT -0.22 bp ($t = -0.10$), Qwen $+4.06$ bp ($t = +1.36$)
  - No significant gain even there: almost all of the reaction happens on day d
- Part of the Qwen edge may be memorised: the Qwen2.5 weights were released on 19 September 2024, after the 2010–2023 headlines
  - A headline that names the episode can trigger recall, unlike the Apple test, which gave only the date

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Sentiment as a Generated Regressor: Classical Error

- The question of this section: what happens to the slope and its interval
  - When the regressor is a model output, not the true tone?
- Model: $r = a + \beta s^* + \varepsilon$ with true tone s^*
  - We observe a model output $\hat{s} = s^* + u$
  - Classical error (u independent of s^* and ε): plim $\hat{b} = \beta\lambda$, $\lambda = \mathrm{Var}(s^*)/[\mathrm{Var}(s^*) + \mathrm{Var}(u)] < 1$
  - λ: the reliability of the score, the share of its variance that is signal
  - The test of $\beta = 0$ stays valid; magnitudes are attenuated towards 0

Standardised Score: Correcting by the Correlation with the True Tone

- Standardised score z, non-differential error: $\text{Cov}(r, z) = \beta \, \text{Cov}(s^*, z)$
  - Non-differential error: $E[r \mid s^*, \hat{s}] = E[r \mid s^*]$
  - Since $\text{Var}(z) = 1$, the slope is $b_z = \text{Cov}(r, z) = \beta \, \sigma_{s^*} \, \rho(s^*, \hat{s})$
  - Slope on z = (effect of one standard deviation of s^*) $\times$ $\rho(s^*, \hat{s})$: divide by the correlation of truth and score
  - A binary label is not classical error: misclassification shrinks a dummy slope by a different factor (Aigner (1973); Appendix and Seminar 15, A1)
- Econometrics of AI-generated variables
  - Battaglia et al. (2024): bias correction, joint estimation
  - Ludwig, Mullainathan & Rambachan (2024): prediction versus estimation uses of LLMs

How Much Attenuation in Our Slopes?

Score	$\rho(s^*, \hat{s})$, Twitter	b on day d (bp)	b/ρ (bp)	ℓ_j, one-factor loading
LM	0.29	15.5	54	0.53
FinBERT	0.58	39.2	67	0.85
Qwen2.5-7B	0.71	41.4	58	0.78

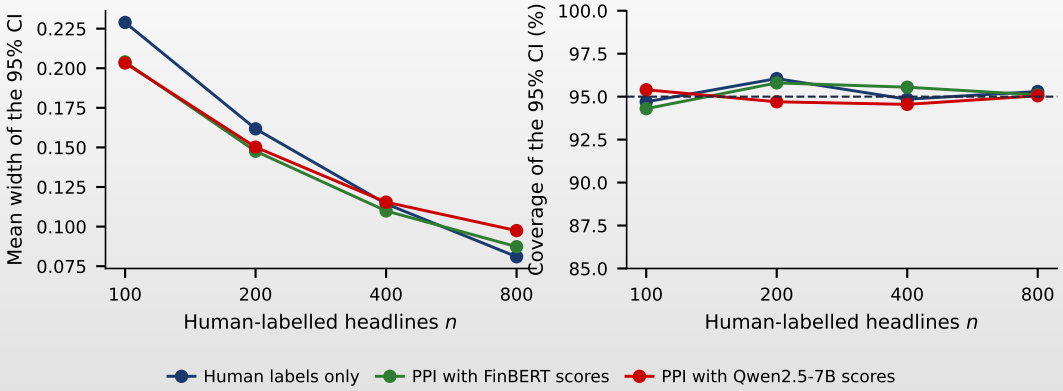
- ρ: correlation of the net score with the human label $(-1, 0, 1)$, one headline at a time
  - b/ρ: same-day effect of one standard deviation of true tone
  - After correction the three methods agree much more: the small LM slope is mostly measurement error, if the assumptions in Further Topics hold
- ℓ_j: loading of score j on the common factor f of the three daily scores
  - $\ell_j = \text{corr}(z_j, f)$ (factor model and instruments: Appendix)
  - Effect of one standard deviation of f, b/ℓ_j: 29 bp (LM), 46 bp (FinBERT), 53 bp (Qwen)

Further Topics: when correction overshoots

Valid Inference with Model Labels: Prediction-Powered Inference

- Data: n texts labelled by humans $(X_j,\ Y_j)$ and N unlabelled texts $\tilde{X}_i$
  - Independent random samples of one population
- Target: $\theta = E[Y]$, the population mean of the human label
  - $f(\cdot)$ is the score of a fixed model, trained on other data
- PPI (prediction-powered inference, Angelopoulos et al. (2023)): $\hat{\theta}_{\mathbf{PPI}} = \frac{1}{N}\sum_{i=1}^{N} f(\tilde{X}_i) - \frac{1}{n}\sum_{j=1}^{n}\left(f(X_j) - Y_j\right)$
  - First term: the mean prediction over the N unlabelled texts
  - Second term: the mean bias $f - Y$, estimated on the labelled texts and subtracted
- $\widehat{\mathrm{Var}}(\hat{\theta}_{\mathbf{PPI}}) = \widehat{\mathrm{Var}}(f)/N + \widehat{\mathrm{Var}}(f - Y)/n$
  - An asymptotically valid CI under this sampling design, whatever the quality of f

PPI on Twitter Headlines: the Bias Correction Is the Main Gain (1/2)



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PPI on Twitter Headlines: the Bias Correction Is the Main Gain (2/2)

- Target: mean human net tone $(-1, 0, 1)$
  - The 2 388 validation headlines act as the population
  - 2,000 replications of n labelled and 2 388 $-n$ unlabelled headlines drawn independently, with replacement
- True mean tone $\theta = 0.054$
  - Plain model means are biased: FinBERT -0.014 (wrong sign), Qwen2.5-7B $+0.101$
  - No sample size fixes this
- PPI keeps nominal coverage (94.3% at $n = 100$)
  - Mean width at $n = 100$: 0.229 with labels only, 0.204 with PPI (FinBERT)
  - Variance reduction $1 - \mathrm{Var}(f - Y)/\mathrm{Var}(Y)$: 24% (FinBERT), 27% (Qwen)
  - Modest because the models are noisy
- At $n = 800$ PPI is wider (0.087 vs 0.081)
  - With only 2 388 $-$ 800 unlabelled texts, $\mathrm{Var}(f)/N$ is no longer negligible
  - For the return regressions, the same labelled sample allows rectifying both OLS moments (Egami et al. (2023))
- What the result supports: comparison of procedures (PPI, human labels only, the mean of the scores)

MFM_ch15_valid_inference

Research Question: Does an LLM Read News as the Market Does?

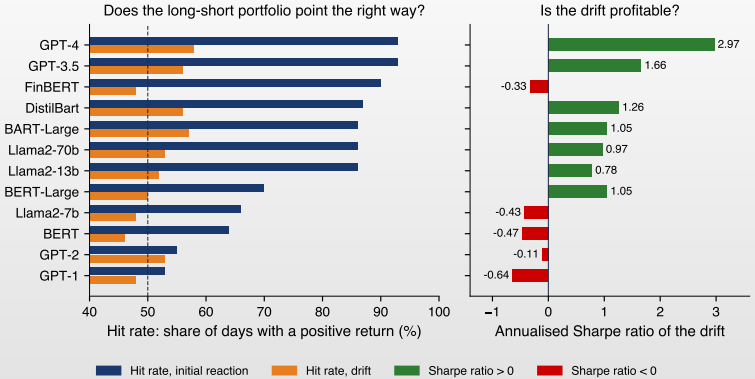
- Lopez-Lira & Tang (2026) (*Journal of Financial Economics*) ask:
  - Can a general-purpose LLM, with no explicit financial training, tell whether a headline is good or bad for the stock?
- Does its answer predict returns?
- Initial reaction: the price move right after the news
  - Not tradable: it measures whether the LLM understands the news
- Drift: the return after the reaction, in the direction of the news
  - Tradable: it measures underreaction of the market
- Versus our FNSPID test: the same zero-shot scoring of headlines
  - But on news after the model's training cutoff and with exact publication times
- Why it matters: the LLM becomes a measuring instrument for how fast prices absorb news
  - Their model: only LLMs above a quality threshold predict returns, and wide adoption should erode the predictability

GPT-4 Gets the Reaction Right and Earns the Drift

Model	Hit rate, reaction	Hit rate, drift	Drift, % per day	*SR*
GPT-4	0.93	0.58	0.34	2.97
GPT-3.5	0.93	0.56	0.29	1.66
FinBERT	0.90	0.48	−0.07	−0.33
GPT-1	0.53	0.48	−0.07	−0.64

- Overnight news, long-short portfolios: GPT-4 gets the direction of the reaction right on 93% of days
  - Like GPT-3.5, but it earns the largest drift
  - Short leg 0.26% per day (*SR* 2.01), long leg 0.08% (*SR* 0.78): the market underreacts mainly to bad news
- Regression, overnight news: $\hat{\gamma} = 1.326$ ($t = 26.52$) for the reaction, 0.161 ($t = 6.59$) for the drift
  - Adding the vendor's sentiment score leaves GPT-4 at 0.158 ($t = 5.27$) and the vendor at 0.010 ($t = 0.15$)
  - Small stocks: interaction coefficient 0.404 ($t = 4.55$): the drift is concentrated where arbitrage is costly
- What the result supports: association
  - The GPT-4 Sharpe ratio falls from 6.54 (2021 Q4) to 1.22 (January–May 2024): Further Topics

Which Models Read the News Right, and Which Earn the Drift (1/2)



BART: Bidirectional and Auto-Regressive Transformer, an encoder–decoder language model; DistilBart: a compressed (distilled) BART; Llama2: open-weight LLMs released by Meta

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Which Models Read the News Right, and Which Earn the Drift (2/2)

- Long-short portfolios on overnight news, twelve models, same design
  - October 2021 – May 2024; source: Lopez-Lira & Tang (2026)
- Threshold: GPT-1, GPT-2, BERT and Llama2-7b get the reaction right on only 53–66% of days
  - They have negative drift Sharpe ratios (-0.64 to -0.11)
  - Seven models reach 86–93%, but only GPT-4 and GPT-3.5 pair it with a large drift Sharpe ratio
- FinBERT gets the reaction right but has a negative drift
  - It gives a direction for only 25% of overnight headlines
  - Reading what is already in the opening price does not earn the drift
- Read the legs: stocks held on average in the short leg
  - 2 stocks for BERT-Large (Sharpe 1.05), 6 for GPT-3.5, 20 for GPT-4
  - A Sharpe ratio from a handful of shorts is fragile

MFM_ch15_llt_case

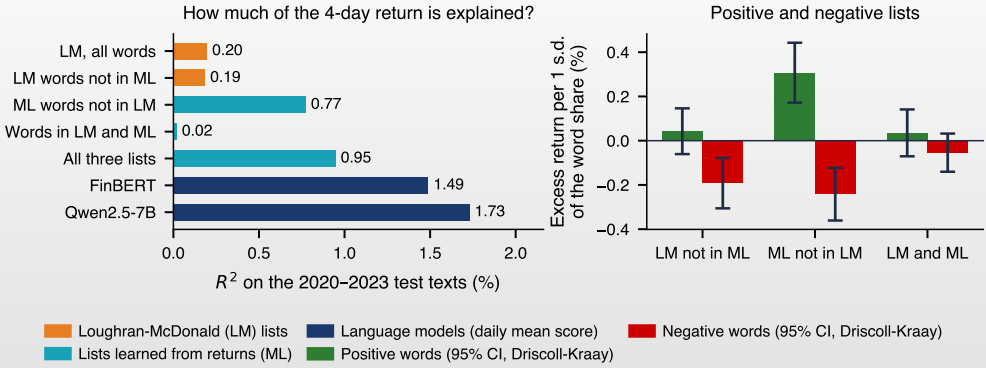
Replication in This Course, Limitations and an Open Extension

- The Qwen2.5-7B strategy of this chapter on 16 FNSPID stocks: +5.64 bp per day
  - Against 34 bp for GPT-4
  - Replication (Seminar 15, C3): the published prompt, temperature 0, scores $\{-1, 0, 1\}$, the two-firm rule and the same regression, on FNSPID headlines and the adjusted daily closes of the course data
  - Our data limits: date-only timestamps (the first safe return is $d + 2$, not the next open), 16 large stocks, headlines that predate the model
- Limits stated by the authors: at 20 bp per round trip the strategy loses money
  - So it suits low-cost traders such as market makers
  - Our 5 bp per one-way trade is 10 bp per round trip
  - Prompts and embedding models were not optimised, so the comparisons are lower bounds
- Open extension, on headlines published after the release of an open-weight model
  - does the drift survive?
  - does its Sharpe ratio keep falling as the authors' model predicts?

A Second Case Study: The Colour of Finance Words

- Garcia, Hu & Rohrer (2023) (*Journal of Financial Economics*), the question:
  - Can a word list chosen by price reactions compete with lists chosen by researchers, such as Loughran–McDonald (LM)?
- The paper's corpus: transcripts of 85 530 earnings calls (the calls in which firms present quarterly results), 3 229 firms, 13 Oct 2005 – 7 Oct 2020
- The label of each text: the stock's return minus the market return, from the close of day $t-1$ to the close of day $t+2$
- The paper's main result: lists learned on the calls before 2016, evaluated on 2016–2020, with the paper's control variables
  - Adjusted R^2: 2.1% with the LM words not in the ML lists, 4.6% with the ML words not in LM
  - With the controls only: 1.7%
  - The lists learned on the paper's full sample share only 18 positive and 30 negative words with LM
- Why it matters: the list stays transparent, but the colour of each word comes from the market, not from the researcher's judgement
  - It is the simple benchmark against which FinBERT and LLMs should be compared on the same texts

Word Lists Learned from Returns, on the FNSPID Headlines



ML (machine learning): the lists learned from returns; the lists are built on 2009–2019, R^2 and slopes are estimated on 2020–2023

MFM_ch15_colour_words ▶ Interactive

Method: Robust MNIR and What the Course Replication Changes

- MNIR (multinomial inverse regression, Taddy (2013), Appendix): the count of word j in text i is Poisson, with mean $m_i e^{\alpha_j + \varphi_j v_i}$
  - m_i: the number of words of the text; v_i: the standardised return
- The robust step: MNIR on 500 subsamples of 5 000 texts each, drawn without replacement
  - D_j^+: the share of subsamples with $\varphi_j > 0$ minus the share with $\varphi_j < 0$
  - Positive list: $D_j^+ \geq 80\%$; negative list: $D_j^+ \leq -80\%$
- The score of a text: the number of list words divided by the number of words, scaled to unit standard deviation
- The course replication, on the chapter's data:
  - The text: all FNSPID headlines of one stock on one day, 18 282 stock-days
  - Training: Oct 2009 – Dec 2019, without the texts whose return window reaches into 2020 (10 286 texts); test: 2020–2023 (7 980 texts)
  - Vocabulary (15 376 words) and winsorization cut-offs: from the training texts only
  - The sign of φ_j: a score-test approximation of the AIC selection (Akaike information criterion, $z^2 > 2$), instead of the gamma-lasso penalty
  - No control variables of the paper; the names of the 16 firms are removed instead of the proper-noun filter

The Result: Words Chosen by the Market Explain More than LM

- R^2 on the 2020–2023 test texts, slopes estimated on the same period: 0.20% with all LM words, 0.77% with the 94 ML words that are not in LM
  - The difference: +0.58 percentage points, 95% CI [0.14; 1.02] (bootstrap over blocks of 10 consecutive days, 1 999 draws, learned lists held fixed)
  - For cut-offs from 50% to 95%, the R^2 of the ML lists is between 0.44% and 1.02%
- One standard deviation more in the share of ML words (t with Driscoll–Kraay standard errors, HAC over days)
  - Positive: +0.31% ($t = 4.45$); negative: −0.24% ($t = -3.98$)
- The coefficient of the positive LM words is not significantly different from zero: +0.04% ($t = 0.81$)
  - Negative LM words: −0.19% ($t = -3.29$), significant also next to the ML lists: −0.17% ($t = -2.90$)
- FinBERT (1.49%) and Qwen2.5-7B (1.73%) explain more than any list
  - Qwen against the ML lists: +0.96 percentage points, 95% CI [0.20; 1.73]
- What the result supports: a comparison of procedures, on an association within the same 4-day window, not a forecast
  - Many ML words describe the price move itself (“jumped”, “slumped”, “gainers”, “losers”): the headline may follow the return
- Limits: only 1 positive and 5 negative words shared with LM (positive; claims, downgrades, weak, worse, worst)
  - Qwen may already know what followed the 2020–2023 news (Look-Ahead Bias)

An Open Question: Can LLM-Generated Hypotheses Be Trusted?

- Ludwig & Mullainathan (2024): hypothesis generation as a statistical procedure
  - A model trained on data reveals a pattern, humans name it, fresh data test it
- Lopez-Lira, Tang & Zhu (2025): before its knowledge cutoff an LLM can recall outcomes
  - So a hypothesis it proposes about that period is not independent of the data that will test it
- Open: when an assistant proposes the signal, the prompt, the threshold and the holding period
  - what is the family of tests?
  - which inference stays valid?
- Agents that run the experiments themselves (Lu et al. (2024)) make the family larger and less visible

The Family of Tests: FWER and FDR

- Formally: $H_{\mathbf{0},j} : E[r_t^{(j)}] = 0$ for candidate strategies $j = 1, \ldots, M$
  - Control the FWER (Holm) or the FDR (Benjamini & Hochberg (1995))
  - FDR: expected share of false rejections among all rejections
  - BH at level q: reject the hypotheses with $p_{(\mathbf{1})}, \ldots, p_{(k^*)}$, where $k^* = \max\{k : p_{(k)} \leq kq/M\}$
  - M counts every variant the assistant proposed, including those discarded; if the hypotheses were built on the test outcomes, the p-values need untouched test data or selective (sequential) inference

The Discovery Loop and What the Researcher Must Check

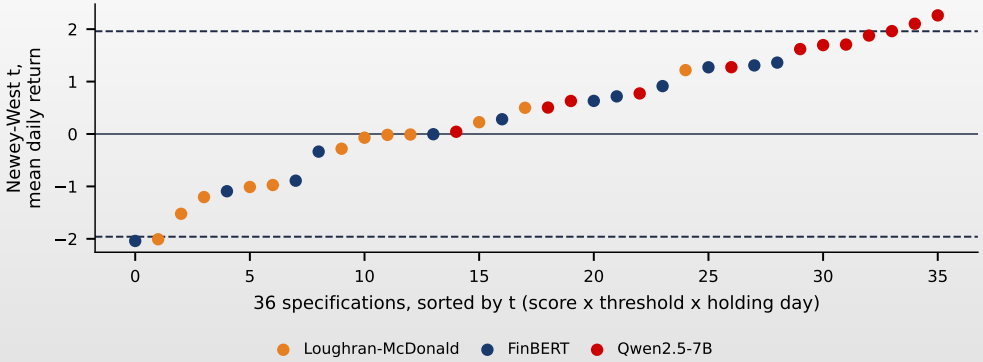
- Literature, hypotheses, data and code, robustness fixed in advance, critique, report
  - An assistant (Claude, ChatGPT, Gemini, GitHub Copilot) drafts each step
  - The researcher verifies each one (reusable prompts: Appendix)
- **Memorisation**: the model knows what followed the headline
  - Test: news after the release of the weights; anonymised text as a diagnostic (Glasserman & Lin (2023)); the memory test of this chapter
- **Forking paths**: prompt, threshold, horizon tuned on the test period
  - Test: a pre-registered grid and a stated error rate: Holm (FWER) or BH (FDR, under independence or positive dependence); $|t| > 3$ (Harvey, Liu & Zhu (2016)) is a heuristic, not an error-rate guarantee
- **HARKing** (hypothesising after the results are known)
  - A plausible mechanism written after the result, with references that may not exist
  - Test: state the mechanism and its falsifier before the test; resolve every DOI (digital object identifier)
- **Timing and agents that run code**
  - Date-only timestamps
  - Silent changes of data, windows or seeds
  - Test: first tradable return on $d + 2$; read-only data, logged tool calls, the diff of every script before a result is accepted

Mini-Case: Refinements Suggested by an Assistant

- A typical assistant proposal:
  - "the Qwen signal works; refine it with a stronger threshold or a later entry day"
  - We test the whole grid on the saved scores
- Grid: 3 scores (LM, FinBERT, Qwen) $\times$ 3 thresholds $\times$ 4 return days = 36 strategies
  - Thresholds $|s_{i,d}| > b$, $b \in \{0, 0.25, 0.5\}$, on the raw score
  - Return days $d+2$ to $d+5$, each a one-day position opened at the previous close
  - Base case: 5.64 bp per day, $t = 2.26$
- $|t| > 1.96$ for 5 of 36 strategies, against 1.8 expected under the global null at 5%
  - 2 of them have the wrong sign
- Survivors: Holm (FWER 5%) 0, BH (FDR 10%) 0, $|t| > 3$ 0
  - The Bonferroni threshold is $|t| > 3.20$
- Reading: the refinements add no evidence
  - The best variant is the one we started from
  - Itself chosen among three scores, on a period the model may remember

MFM_ch15_ai_discovery

The Specification Curve: Is It Different from Noise? (1/2)



MFM_ch15_valid_inference ▶ Interactive

The Specification Curve: Is It Different from Noise? (2/2)

- All 36 strategies of the mini-case sorted by t (Simonsohn, Simmons & Nelson (2020)); dashed: ± 1.96
- Joint null: no specification predicts returns
  - Flip the signs of the positions at random in blocks of 10 consecutive trading days, zero-return days included, 2 000 times, with the same flips for all 36 specifications (dependent wild bootstrap, Shao (2010))
  - Valid if, under the null, signed returns are symmetric around 0 and blocks are roughly independent
- Observed: 3 specifications with $t > 1.96$, median $t = 0.50$
  - Under the null, 3 or more occur with probability 0.113; a median this high with probability 0.207
- Reading: the curve is compatible with noise at 5%
  - Reporting only the best point would hide this
- What the result supports: association
  - None of the 36 predictive associations stands out from noise

MFM_ch15_valid_inference

Your Turn: a Project Seed

- Question: does LLM headline sentiment predict returns beyond $d + 1$?
  - On news published after the model's release
  - Under a pre-registered grid
- Replicate first the Qwen strategy of this chapter, then the post-cutoff design of Lopez-Lira & Tang (2026)
- Deliverables
  - The pre-registered grid and correction; dated headlines after the release of an open model; the test on all variants
  - An LLM-proposed mechanism, stated before the test with its falsifier; a referee-style critique and your reply
- Log each prompt that shaped a decision in `AI_USE.md`
  - Log each invented reference, memorised fact or wrong number caught in `AI_ERRORS.md`, with the test that caught it (templates)

Key Takeaways

- Generic dictionaries misread finance
  - The LM dictionary is a transparent baseline
- FinBERT and 7B–14B open LLMs show no significant accuracy difference on financial headlines
  - Not proof of equality
  - A few hundred labels and a simple classifier compete with both
- LLMs remember market history
  - Evaluate them only after their release date
- Headline sentiment explains same-day moves
  - With realistic timing and costs it does not beat the market in the 16 large stocks studied
- Scores are measurements with error
  - Attenuation, possibly shared errors across models, few clusters and many tries all shape the inference

Check your understanding

- Why can a method that adds up the dictionary labels of isolated words misclassify a financial phrase?
  - Give an example in which context changes the sign.
- Can a high classification accuracy establish trading profitability?
  - Separate label quality, timing and transaction costs.
- Why must a historical news backtest consider the model's release date?
  - Describe the data choice that rules out the evaluated returns being memorised in the weights.
- Why is a sentiment score a measurement rather than an observed truth?
  - Explain what a random sample of human-labelled headlines allows you to estimate.

Short model answers: Appendix.

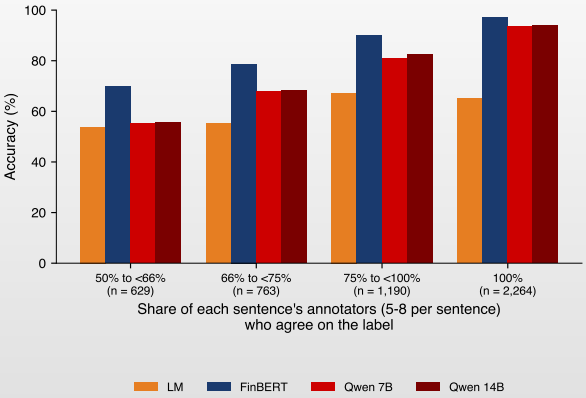
The Practical Decision

- **What I choose**
  - A fast, transparent baseline: the LM dictionary, no training and auditable
  - Short English financial headlines: FinBERT or, with a few hundred labels from the domain, embeddings + logistic regression
  - No labels, other languages or long documents: an open LLM of at least 7B, zero-shot
  - For daily trading of the 16 large stocks: no sentiment signal
- **Under which conditions**
  - Out-of-sample evaluation, with intervals and paired tests
  - On Twitter, TF-IDF + logistic regression reaches 83.2% accuracy, against 74.2% for Qwen2.5-7B and 72.5% for FinBERT
  - The Qwen2.5-7B strategy breaks even at 1.5 bp per trade, below the assumed cost of 5 bp
- **What result would make me drop it**
  - A significant McNemar test in favour of another method, on texts from the target domain
  - A memory test with an AUC above 0.5 over the backtest period (0.69 for Qwen2.5-14B on the months before release)
  - A signal that stays significant after Holm on headlines published after the model's release, with a break-even cost above the real cost

Worked Example: One Sentence, Two Dictionaries

- Financial PhraseBank sentence, labelled **positive** by all annotators:
  - *Uponor improved its performance in spite of the decrease in residential construction in the US .*
- LM: positive words improved; negative words –
  - $\tau = (1-0)/(1+0) = +1.00$: positive, correct
- GI: positive words –; negative words spite, decrease
  - $\tau = (0-2)/(0+2) = -1.00$: negative, wrong
  - "Spite" and "decrease" describe the market, not the firm, and "improved" is not in GI

When Humans Disagree, Models Disagree Too



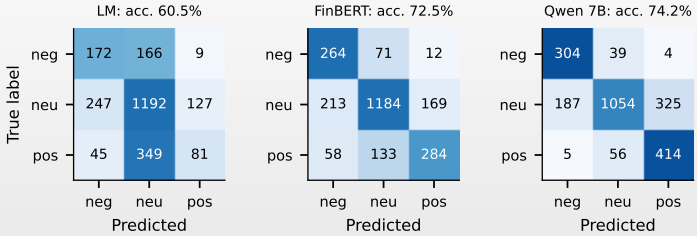
- Financial PhraseBank split by the highest agreement threshold each sentence reaches
  - Lowest group: 50% to below 66%
- Full agreement: FinBERT 97%, Qwen2.5-14B 94%, LM 65%
  - Agreement from 50% to below 66%: FinBERT 70%, Qwen2.5-14B 55%, LM 54%
- The human label is itself noisy: it caps measured accuracy
  - Latent-label models: Appendix
  - Symmetric noise, independent of the classifiers, shrinks measured differences towards 0; class-dependent noise can enlarge or reverse them
- For signals: a strong, unambiguous tone is more informative than a borderline one

MFM_ch15_classification

Agreement beyond Chance: Cohen's κ

- Accuracy also counts agreement reached by chance, which is large when one class dominates
- Cohen's κ (Cohen (1960)): $\kappa = \dfrac{p_o - p_e}{1 - p_e}$
  - p_o: the share of texts on which two raters (here the model and the human) give the same label
  - $p_e = \sum_k p_k^A \, p_k^B$: the agreement expected by chance, from the marginal label frequencies of the two raters
  - $\kappa = 1$: perfect agreement; $\kappa = 0$: no better than chance
- Twitter validation set: $\kappa = 0.19$ (LM), 0.50 (FinBERT), 0.56 (Qwen2.5-7B)
- The "always neutral" benchmark has $\kappa = 0$, although its accuracy beats that of LM
- For more raters per text: Fleiss' κ (Appendix)

Where the Errors Are: Different Models, Different Mistakes



- Twitter validation set: rows are the true labels, columns the predicted labels
  - Counts in the cells
- LM sends most positive headlines to neutral: its positive list is short
- FinBERT errs in both directions around neutral
- Qwen2.5-7B: recall of about 87% on positive (414/475) and 88% on negative headlines (304/347)
  - But precision of only 56% on positive (414/743) and 61% on negative (304/496)
  - Many neutral headlines are called positive or negative
- For trading, false positives cost money: precision on the extreme classes matters

MFM_ch15_classification

Same Model, Different Wording: the Prompt Is a Parameter



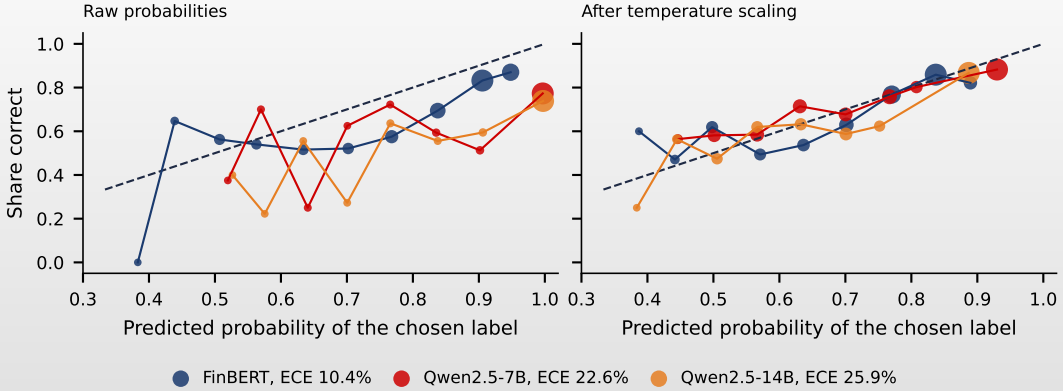
- Three prompts that ask the same question; Twitter validation set
- Accuracy ranges: 1.5B from 52.7% to 61.7%; 7B from 73.7% to 77.8%; 14B from 69.9% to 75.6%
  - The three prompts give the same label on only 67% of headlines (7B)
  - Share of "neutral" answers for 7B: 48% (P1) to 69% (P3)
- Choosing the prompt after seeing test results is a form of data snooping (Chapter 13)
  - Fix the prompt in advance, or choose it on a separate validation set

MFM_ch15_llm_zero_shot

Calibration of Probabilities: ECE and Temperature Scaling

- The question: when the model gives a class probability 0.9, is it right 90% of the time?
- ECE (expected calibration error) $= \sum_b \frac{n_b}{n} |\text{acc}_b - \text{conf}_b|$ over 10 bins of the top probability
  - n_b and n: the number of headlines in bin b and the total number of headlines
  - conf_b: the mean top probability in bin b
  - acc_b: the share of headlines classified correctly in bin b
- Temperature scaling (Guo et al. (2017)): $p_k(T) \propto p_k^{1/T}$, i.e. logits divided by T
  - T fitted by maximum likelihood on one random half of the Twitter validation set
  - ECE measured on the other half
  - The saved probabilities are rounded to 6 decimals; zeros are replaced numerically by 10^{-12}
  - The most likely class does not change, hence neither does accuracy

Are the Probabilities Calibrated? LLMs Are Overconfident (1/3)



MFM_ch15_valid_inference

Are the Probabilities Calibrated? LLMs Are Overconfident (2/3)

- Left panel: raw probabilities; right panel: after temperature scaling
  - The diagonal means perfect calibration
- ECE of the raw probabilities: FinBERT 10.4%, Qwen2.5-7B 22.6%, Qwen2.5-14B 25.9%
- Qwen2.5-7B gives 88% of headlines a confidence above 0.93 (mean 99.7%)
  - It is right on only 77% of them: overconfidence
- Temperature scaling, $T = 8.4$ (7B) and 1.4 (FinBERT): ECE falls to 4.3% and 4.6%
  - T for 7B depends on the numerical floor: 5.6 with 10^{-6} instead of 10^{-12}; the overconfidence remains
- ECE of 7B depends on the prompt: 22.6%, 22.1%, 19.8% under P1, P2, P3

MFM_ch15_valid_inference

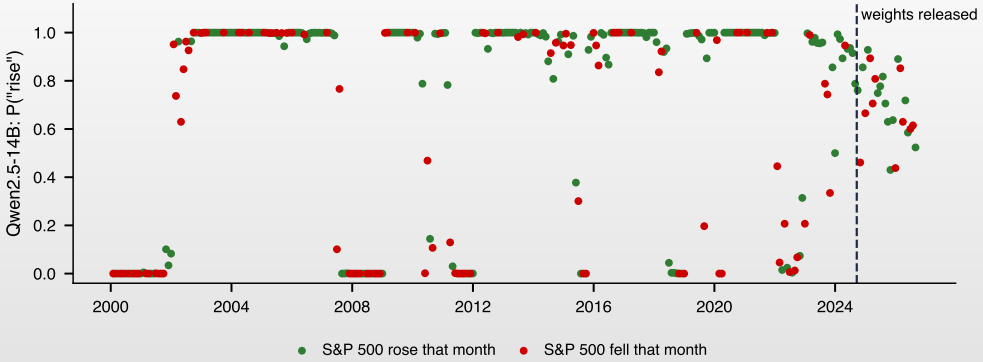
Are the Probabilities Calibrated? LLMs Are Overconfident (3/3)

- Two proper scores of the probabilities, lower is better
  - log loss: the mean of $-\ln p_{iy_i}$, with p_{iy_i} the probability given to the true class y_i of headline i
  - Brier score (Brier (1950)): squared distance between the probabilities and the one-hot label (1 for the true class, 0 otherwise), summed over the 3 classes
- For 7B, temperature scaling moves the log loss from 2.61 to 0.63 and the Brier score from 0.468 to 0.366
- With raw LLM probabilities the net score in a regression is close to a three-valued label

Worked Example: LLM versus FinBERT

- Twitter, prompt P1: Qwen2.5-7B 74.2%, FinBERT 72.5%
  - A = FinBERT, B = Qwen; texts that only one model gets right: $n_{01} = 395$, $n_{10} = 355$
  - Gap $(n_{01} - n_{10})/n = 40/2\,388 = +1.7$ points, CI 95% $[-0.5; +3.8]$; by hand, $\chi^2 = (|n_{01} - n_{10}| - 1)^2/(n_{01} + n_{10}) = 2.03$, so $p = P(\chi^2_1 > 2.03) \approx 0.15$ (exact test: 0.154)
  - Qwen2.5-14B against FinBERT: -0.1 points, $p = 0.942$
- No significant difference, which is not proof of equal accuracy
  - 70 to 130 times more parameters and slower processing bring no demonstrated gain
  - Where LLMs help: no labels needed, any language, longer and more complex documents (Appendix)

What Qwen2.5-14B "Remembers" About the S&P 500 (1/2)



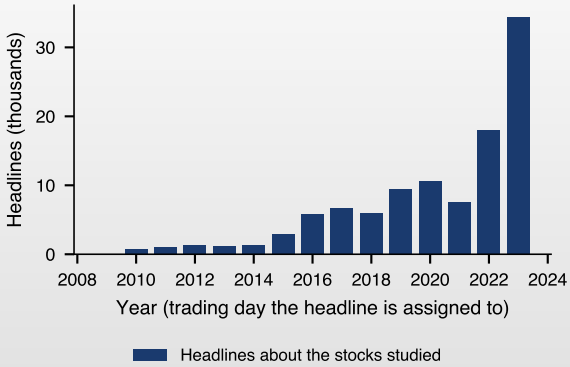
MFM_ch15_lookahead

What Qwen2.5-14B "Remembers" About the S&P 500 (2/2)

- Probability of the answer "rise" for each month, no context given
  - Colour: what the index actually did
- "Fall" for 2000–2002 and 2008–2009, "rise" for most of 2003–2007 and 2013–2021
  - October 2008 (-16.9%), March 2020 (-12.5%), December 2018 (-9.2%): $P(\text{rise}) \approx 0$
- But September 2002 (-11.0%) and February 2009 (-11.0%) get $P(\text{rise})$ close to 1
  - Knowledge is coarse: memorable episodes rather than every month
- After the release the answers drift towards "rise"
  - No knowledge, only the general belief that markets go up

MFM_ch15_lookahead

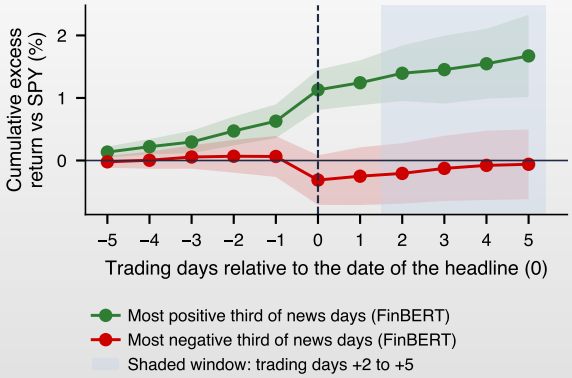
How Many Headlines, and When: Coverage Is Uneven



- Headlines about the 16 stocks per year, by the trading day they are assigned to
- 34 380 headlines in 2023 alone, against 8 445 for all years before 2016
  - Results are dominated by recent years; early years have few news days per stock
- On a news day a stock has on average 5.8 headlines
  - Daily score: the average of the headline scores for that stock and day
- Survivorship bias: the 16 stocks belong to today's large firms, selected with hindsight

MFM_ch15_news_data

Event Study: the Move Happens Before and On the News (1/2)



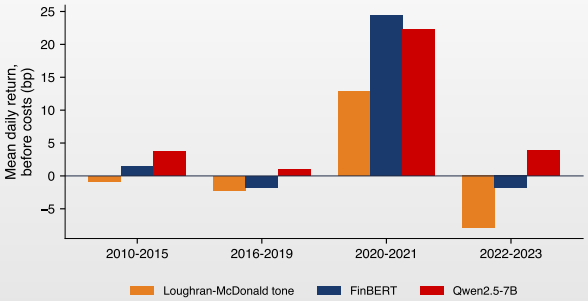
- Mean cumulative excess return, days -5 to $+5$
  - News days split by terciles of the daily FinBERT score
  - Shaded: tradable days $d+2$ to $d+5$
- Most positive third: $+0.63\%$ over the 5 days before, $+0.50\%$ on day d
  - Most negative third: $+0.06\%$ and -0.38%
  - Leaks, or headlines written after the price moved

MFM_ch15_news_signal

Event Study: the Move Happens Before and On the News (2/2)

- Sentiment effect = positive minus negative
  - Estimand: the difference in means, each stock-day event weighted equally; Driscoll–Kraay CI (HAC over days)
  - Day d: $+1.00\%$ $[+0.87; +1.13]$; days $d+2$ to $d+5$: $+0.17\%$ $[-0.11; +0.46]$
  - The chart averages within each day first, then over the days of each third: $+0.88\%$ on day d
- Test robust to event-induced variance (Boehmer, Musumeci & Poulsen (1991)), $d+2$ to $d+5$
  - $z = \overline{SCAR}/(s(SCAR)/\sqrt{N})$, with $SCAR_e = CAR_e/\hat{\sigma}_e$ the cumulative excess return of event e divided by its standard deviation estimated before the event; $N = 6\,094$ events per third
  - $z = +2.62$ (positive), $+1.25$ (negative) if events are independent; with Driscoll–Kraay standard errors: $+1.57$ and $+0.66$
  - Both thirds drift up; the selection of today's large firms may contribute

Stability over Time: Weak Evidence



- Mean daily return of the strategy before costs, by subperiod
- FinBERT: 2010–2015 +1.4 bp, 2016–2019 −1.8 bp, 2020–2021 +24.3 bp, 2022–2023 −1.8 bp
  - Only 2020–2021 is large, yet not significant at 5%: FinBERT $t = +1.89$, Qwen +22.2 bp ($t = +1.78$)
  - 2020–2021: the meme-stock period (GameStop episode: Appendix)
  - After the assumed costs the point estimates stay positive, but the statistical evidence is weak
  - Over the full sample the net result is negative
- Why so little: 16 heavily followed stocks, date-only timestamps, daily frequency
  - Published predictability relies on precise timestamps and many small stocks (case study)

MFM_ch15_news_signal

When Does the Correction Overshoot? Headlines versus Stock-Days

- The correction b/ρ rests on three assumptions
  - Non-differential error: given the true tone, the score error does not predict returns
  - The Twitter correlation carries over to FNSPID headlines
  - The same unit: ρ is measured per headline, but b uses the daily score of a stock, the mean of on average 5.8 headlines
- If headline errors are independent and the headlines of a stock-day share one true tone:
  - Averaging raises the correlation with the truth
  - The daily ρ exceeds the headline ρ, and b/ρ over-corrects
- Numerical example: one headline has reliability $\lambda = \rho^2 = 0.25$ ($\rho = 0.50$)
  - The mean of $k = 4$ such headlines has $\lambda_k = \dfrac{k\lambda}{1 + (k-1)\lambda} = \dfrac{4(0.25)}{1 + 3(0.25)} = 0.57$ (Spearman–Brown)
  - So $\rho_k = 0.76$
  - A daily slope of 10 bp is then corrected to $10/0.50 = 20$ bp with the headline ρ, but to $10/0.76 \approx 13$ bp with the daily one
- With independent headline errors our b/ρ values are upper bounds
  - The conclusion about the LM slope depends on this
- Remedy: labelling every headline of a random sample of stock-days measures ρ at the right level

When PPI Helps, and How It Extends to Regressions

- The PPI interval is narrower than labels alone when $\mathrm{Var}(f)/N + \mathrm{Var}(f - Y)/n < \mathrm{Var}(Y)/n$
  - Roughly $\mathrm{Var}(f - Y) < \mathrm{Var}(Y)$ when $N \gg n$: the model must explain part of the label's variation
- For a regression the rectifier applies to every moment condition that contains the predicted variable
  - Also when it is a regressor (Egami et al. (2023))
- Egami et al. (2023): design-based supervised learning
  - A doubly robust version that allows labelling probabilities to depend on the text
- The few hundred labelled documents are not only training data
  - They are the validation sample that makes downstream inference valid

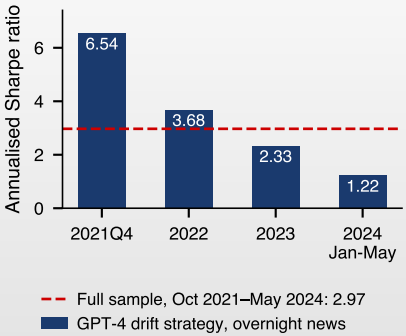
How the Authors Score News Published after the Training Cutoff of GPT-4

- 159,137 firm-headline-date observations, 4,123 US common stocks
  - Listed on the NYSE, NASDAQ and AMEX exchanges
  - October 2021 – May 2024
  - The sample starts after the September 2021 training cutoff of the scored model, `gpt-4-0314`
  - Filters: news-vendor relevance score 100, no "stock-gain"/"stock-loss" headlines, only new events, near-duplicates of the same day removed
  - Overnight news (before 9 a.m. or after 4 p.m.): 82% of headlines; the rest are intraday
- Prompt: the model acts as a financial expert
  - It answers `YES` (good news), `NO` (bad news) or `UNKNOWN` on the short-term effect on the stock price
  - Temperature 0
  - Score $s_{i,t}$: `YES` $= 1$, `UNKNOWN` $= 0$, `NO` $= -1$

Portfolios and Regression in Lopez-Lira and Tang

- Portfolios: each day, equal weights
  - Long the stocks with a positive score and short those with a negative one
  - A leg only with at least two firms
  - For overnight news, initial reaction: previous close to open; drift: open to close
  - Hit rate: share of days with a positive portfolio return; Sharpe ratio annualised from daily portfolio returns, as for our strategy
- Regression: $r_{i,t} = a_i + b_t + \gamma\, s_{i,t} + \varepsilon_{i,t}$
  - Firm and date fixed effects a_i, b_t
  - Standard errors clustered by date and by firm
  - γ: return per unit of score
  - Twelve language models of increasing size scored with the same design, from GPT-1 to GPT-4

The Drift Fades as LLMs Spread



- GPT-4 long-short drift strategy on overnight news, annualised Sharpe ratio by subperiod (source: Lopez-Lira & Tang (2026))
- From 6.54 in 2021Q4 to 1.22 in January–May 2024, more than a fivefold drop
  - The full-sample 2.97 pools a strong start with a weak end: not a forecast of today's performance
- Consistent with the erosion predicted by the authors' model, and the timing matches the spread of ChatGPT
- The authors also name changing market conditions
  - Short subperiods: the 95% intervals are wide, and the one for 2024 includes 0
  - Intraday news: the close-to-close drift is no longer significant in 2024

MFM_ch15_llt_case

When Annotators Disagree: Agreement and Latent Labels

- **Fleiss' κ** (Fleiss (1971)): N texts, m raters, n_{ij} raters giving text i label j
  - $\bar{P} = \frac{1}{N}\sum_i \sum_j \frac{n_{ij}(n_{ij}-1)}{m(m-1)}$, $\bar{P}_e = \sum_j p_j^2$, $\kappa = \dfrac{\bar{P} - \bar{P}_e}{1 - \bar{P}_e}$; p_j: share of all votes for j
  - The formula assumes the same m for every text; PhraseBank has 5–8 raters per sentence and publishes only agreement levels, not the votes: κ cannot be computed from the release
- **Dawid–Skene model** (Dawid & Skene (1979)): rater a has a confusion matrix
  - $\pi_{jl}^{(a)} = P(\text{says } l \mid \text{true } j)$
  - The true class is latent
  - EM (expectation–maximisation) estimates the $\pi^{(a)}$ and gives each text a soft label $P(\text{true} = j \mid \text{votes})$
- Evaluate models by a proper scoring rule (log loss) against soft labels
  - Not by accuracy against the majority label
  - The ceiling in the agreement chart is a label-noise ceiling, not a model failure

Back

Worked Example: Is FinBERT Better than LM?

- Twitter validation set, 2 388 headlines: LM 60.5%, FinBERT 72.5%
  - LM wrong and FinBERT right: $n_{01} = 539$; LM right and FinBERT wrong: $n_{10} = 252$
  - Exact McNemar $p < 0.001$
- Difference +12.0 percentage points, bootstrap CI 95% [+9.8; +14.2]
  - Clear win for the fine-tuned model, even outside its training data

Back

Processing Time of the Qwen2.5 Models

- Wall-clock time per headline on a laptop graphics processor, including the prompt
  - Median of one pass per prompt
  - 0.5B: 4 ms; 1.5B: 12 ms; 3B: 20 ms; 7B: 92 ms; 14B: 105 ms
  - Single passes are noisy: 7B took 51 to 94 ms, 14B 88 to 112 ms; a timing claim needs repeated runs

Back

Beyond Headlines: Long Documents

- Annual reports: Loughran & McDonald (2011) built their lists from 10-K filings
  - Tone and uncertainty predict returns and volatility around filings
- Earnings calls and analyst reports: Huang, Wang & Yang (2023) show FinBERT extracts information that dictionaries miss
- Kim, Muhn & Nikolaev (2023): LLM summaries of disclosures are much shorter
  - But they explain stock-market reactions better than the originals
  - They measure disclosure "bloat": long, uninformative text is linked to lower price efficiency
- Long documents need models with long context windows, or chunking and aggregation

Back

Text Regressions with Statistical Theory

- Taddy (2013): multinomial inverse regression: model word counts given the outcome
  - $c_d \sim \text{MN}(q(y_d), m_d)$ (MN: the multinomial distribution)
  - c_d: counts, $m_d = \sum_j c_{dj}$: length
  - Log-linear model $q_j(y) \propto e^{\alpha_j + \varphi_j^\top y}$, loadings $\Phi = (\varphi_1, \ldots, \varphi_p)^\top$: given m_d and this model, $z_d = \Phi^\top c_d / m_d$ is sufficient for y_d
- Ke, Kelly & Xiu (2019) (SESTM, sentiment extraction via screening and topic modelling)
  - Screen words correlated with returns
  - Fit a two-topic model of positive and negative words
  - Score new articles by penalised likelihood
  - The paper gives statistical guarantees for the estimator; the signal is strongest on the day of and the day after the news
- Bybee et al. (2024): news attention to topics in 800,000 Wall Street Journal articles
  - It tracks economic activity and forecasts aggregate stock returns
  - Also Jegadeesh & Wu (2013) (word weights from 10-K reactions); rule for all: estimate on data before the forecast period

Back

Did the Memory Vanish? A Formal Test and Its Power

- H_0 : $\text{AUC}_{\text{before}} = \text{AUC}_{\text{after}}$, two independent samples
  - DeLong variance for each AUC (DeLong et al. (1988))
  - $z = (\widehat{\text{AUC}}_1 - \widehat{\text{AUC}}_2)/\sqrt{\hat{V}_1 + \hat{V}_2}$: 14B $z = 0.73$ ($p = 0.47$), 7B $z = 1.01$ ($p = 0.31$)
- Under H_0 (AUC $= 0.5$, no ties): $SE_0 = \sqrt{\dfrac{n_1 + n_0 + 1}{12\, n_1 n_0}}$
  - n_1, n_0: the numbers of up and down months
  - After release: $n_1 = 13$, $n_0 = 10$, so $SE_0 = \sqrt{24/1\,560} = 0.124$
- Smallest detectable drop (5% two-sided test, 80% power):
  - $2.80\sqrt{SE_{\text{before}}^2 + SE_0^2} = 2.80\sqrt{0.03227^2 + 0.124^2} = 0.36$
  - $2.80 = 1.96 + 0.84$, the normal quantiles for the test and for the power; SE_{before}: DeLong standard error of the 296 pre-release months
- A fall from 0.69 to 0.5 (0.19) is below this threshold
  - Not rejecting H_0 does not show that the memory survives release
- Months needed: the smallest m for which the detectable drop falls to 0.19
  - With the post-release share of up months kept at 13/23: $m = 101$ (7B: 129)

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Discrete Labels: Misclassification Is Not Classical Error

- Binary "negative news" dummy $s^* \in \{0, 1\}$, predicted label $\hat{s}$
  - $\alpha_0 = P(\hat{s} = 1 \mid s^* = 0)$, $\alpha_1 = P(\hat{s} = 0 \mid s^* = 1)$
  - The error $u = \hat{s} - s^*$ is negatively correlated with s^*: the classical formula does not apply (Aigner (1973))
  - plim $\hat{b} = \beta \dfrac{\pi(1-\pi)(1-\alpha_0-\alpha_1)}{p(1-p)}$, $\pi = P(s^* = 1)$, $p = P(\hat{s} = 1)$
- Twitter validation set, $\pi = 14.5\%$, from the confusion matrices
  - FinBERT: $\alpha_0 = 13.3\%$, $\alpha_1 = 23.9\%$, factor 0.45; Qwen2.5-7B: 0.59; LM ($\alpha_1 = 50.4\%$): 0.28
- As long as $\alpha_0 + \alpha_1 < 1$ the sign is kept, but a dummy slope can shrink to a third or less

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Two Noisy Measures: Factor Model and Instruments

- One factor: $z_j = \ell_j f + u_j$, $\mathrm{Var}(f) = 1$
  - u_j uncorrelated with f and with each other
  - Then $\rho_{jk} = \ell_j \ell_k$, so $\ell_j^2 = \rho_{jk}\rho_{jl}/\rho_{kl}$
  - Daily FNSPID scores: $\ell_j = 0.53$ (LM), 0.85 (FinBERT), 0.78 (Qwen)
- IV (instrumental variables): with $r = a + \beta f + \varepsilon$, instrument z_Q with z_F
  - $\hat{b}_{IV} = \widehat{\mathrm{Cov}}(r, z_F)/\widehat{\mathrm{Cov}}(z_Q, z_F) \to \beta/\ell_Q$
  - If $\ell_F \neq 0$, f is uncorrelated with ε, and u_F is uncorrelated with u_Q and ε
  - Day d, Qwen: 59 bp [53; 66] with FinBERT as instrument, 37 bp [28; 46] with LM (bootstrap over days)
- If both instruments were valid the two estimates would coincide
  - Difference 22 bp, $z = 5.4$: rejected
  - The test rejects the joint restrictions, not a named instrument: FinBERT–Qwen errors correlated through overlapping training text are one plausible cause, an LM error related to returns another; the factor loadings ℓ_j are then suspect too

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Social Media and Crowds: GameStop, 2021

- Bollen, Mao & Zeng (2011): collective Twitter mood predicted the Dow Jones in 2008
  - The result is sensitive to the sample and method
- January 2021: retail investors coordinated on social media
  - They pushed GameStop up by more than ten times in a few weeks
  - Attention, sentiment and trading rose together; this episode alone cannot separate prediction from causation: each needs its own test
- Lessons for text signals
  - Volume of attention can matter more than tone
  - Crowded signals revert and create risk for short sellers (Chapter 17: bubbles)



A GameStop store, Vallejo, California, February 2021

Photo: Will Buckner (2021); CC BY 2.0; Wikimedia Commons

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The Discovery Loop, Reusable Prompts (1/2)

- **Literature**
  - `"Find studies since 2023 that predict stock returns with LLM sentiment and test after the model's knowledge cutoff: DOI or arXiv id, cutoff date, test window, headline estimate."`
  - Resolve every identifier
  - Semantic Scholar, Google Scholar, Elicit for the search
  - Lopez-Lira & Tang (2026) is the reference design
- **Hypotheses**
  - `"Propose mechanisms by which headline sentiment could predict returns after d+1 and, for each, the cross-sectional pattern that would falsify it."`
  - Under-reaction predicts drift concentrated in small, less-covered stocks
  - Memorisation predicts an edge that disappears after the release date
- **Data and code**
  - `"From the daily news scores of this chapter and the daily closes of the course, reproduce the daily Qwen strategy of this chapter (enter at the close of d+1, hedge with SPY) and its Newey-West t."`
  - The scores are saved, so no model needs to be re-run

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The Discovery Loop, Reusable Prompts (2/2)

- **Robustness**, fixed in advance
  - `"Before any new run, write the full grid of variants you would test and the multiple-testing correction; all of them will be reported."`
  - Placebo: scores shuffled across stocks within the same day
  - Subperiods
  - Costs of 5 bp per trade
- **Critique**
  - `"Write a referee report: every way this strategy could reflect look-ahead, memorisation, selection or timestamps, with the test for each."`
  - Then check each proposed test yourself: a critique can also contain invented results
- **Report**: the whole grid, the variant chosen before the results, model versions and release dates

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AI Agents in Trading: Uses and Risks

- An agent is an LLM in a loop: read inputs, decide, call tools, observe, repeat
  - Tools: data, code, news search, order entry through an API, application programming interface
  - Used today for research summaries, coding and monitoring; fully autonomous trading is still rare and experimental
- Manipulation: a text read by the agent can carry instructions (prompt injection)
  - E.g. a fake press release
- Systemic risk (International Monetary Fund, IMF (2024)): many firms with similar models may trade alike
  - This raises the speed and correlation of moves in stress
  - Controls: position limits, human approval of orders, logging of every step, tests after each model update

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Pitfalls Checklist

- **Training overlap**: scored on its own training data, a model looks better than it is
  - FinBERT on PhraseBank
- **Look-ahead**: an LLM may remember the market after the news
  - Test after its release date
- **Timing**: date-only timestamps
  - Trade no earlier than the close after the news is surely public
- **Snooping**: prompts, thresholds and models chosen after seeing the test results
- **Costs and crowding**: small daily edges vanish after a few basis points of costs
- **Generated regressors**: scores are estimates
  - Correct attenuation
  - Validate on a random human-labelled sample (PPI)

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Short Answers to the Check Questions (1/2)

- **Dictionary**: a method that adds up isolated-word labels gives each word a single label
  - In "lower costs" it applies the negative label of "lower" without interpreting the phrase
  - In the usual context of firm profitability, lower costs can be favourable
  - "capital" or "board" are neutral business nouns that GI counts as negative
- **Accuracy and profit**: accuracy is measured against noisy human labels on other texts
  - A trade needs the first safe return $(d + 2)$
  - It must cover $4c$ a day and must survive the correction for many tries
  - Here all three scores explain same-day moves, yet no strategy earns a net profit after costs

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Short Answers to the Check Questions (2/2)

- **Release date**: before it, the evaluated returns may already be memorised in the weights
  - News and returns after the weight release could not have been memorised in them
  - Condition: no tool access to future information
  - The other sources of look-ahead bias in the backtest still need to be controlled
- **No-context memory test**: it can flag memorisation
  - Qwen2.5-14B: AUC 0.69 on the months before the release
  - A non-significant post-release AUC (0.59 on 23 months) does not establish the absence of contamination
- **Score as a measurement**: $\hat{s} = s^* + u$ estimates the true tone
  - Even a classical, zero-mean error can attenuate the slope (in the classical model, slope on z = effect $\times$ ρ)
  - A random human-labelled sample allows the mean bias $f - Y$ and the reliability of the score to be estimated
  - The estimates hold for the unit of analysis at which the sample was measured
  - PPI corrects the mean score using the mean difference $f - Y$
  - The b/ρ correction requires the model assumptions and a ρ measured at the same unit of aggregation as the regression
  - A ρ measured on headlines does not automatically justify correcting a slope on daily averages

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Text Analysis: Readings and Applications

- Tetlock (2007): media pessimism predicts market returns
- Loughran & McDonald (2011): a finance-specific sentiment dictionary
- Gentzkow, Kelly & Taddy (2019): text as data in economics
- Crypto applications
  - Crypto-asset classification Pele et al. (2023)
  - Dogecoin bubbles and sentiment Pele (2025)
  - Sentiment Quantlet Pele (2024)
- Further study: Quantinar: Natural Language Processing, Quantinar: DELTA, Quantinar: LDA Extensions
  - Textbook Franke, Härdle & Hafner (2019) for the statistics behind the tests

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Acronyms used in this material (1/2)

- **AI**: Artificial Intelligence
- **AIC**: Akaike Information Criterion
- **AMEX**: American Stock Exchange
- **API**: Application Programming Interface
- **AUC**: Area Under the (ROC) Curve
- **BART**: Bidirectional and Auto-Regressive Transformer
- **BERT**: Bidirectional Encoder Representations from Transformers
- **BH**: Benjamini–Hochberg (procedure)
- **BY-NC**: Creative Commons Attribution-NonCommercial
- **CI**: Confidence Interval
- **CLS**: CLaSsification token (BERT-type models)
- **DOI**: Digital Object Identifier
- **ECE**: Expected Calibration Error
- **EM**: Expectation-Maximisation (algorithm)
- **FDR**: False Discovery Rate
- **FNSPID**: Financial News and Stock Price Integration Dataset
- **FWER**: Family-Wise Error Rate
- **GI**: (Harvard) General Inquirer, dictionary IV-4
- **GPT**: Generative Pre-trained Transformer

Acronyms used in this material (2/2)

- **HARKing**: Hypothesising After the Results are Known
- **IMF**: International Monetary Fund
- **IV**: Instrumental Variables
- **LLM**: Large Language Model
- **LM**: Loughran-McDonald (financial sentiment dictionary)
- **LR**: Logistic Regression
- **MIT**: Massachusetts Institute of Technology
- **ML**: Machine Learning
- **MN**: MultiNomial (distribution)
- **MNIR**: MultiNomial Inverse Regression
- **NASDAQ**: National Association of Securities Dealers Automated Quotations
- **NYSE**: New York Stock Exchange
- **OLS**: Ordinary Least Squares
- **PCA**: Principal Component Analysis
- **PPI**: Prediction-Powered Inference
- **RLHF**: Reinforcement Learning from Human Feedback
- **ROC**: Receiver Operating Characteristic (curve)
- **SE**: Standard Error
- **SESTM**: Sentiment Extraction via Screening and Topic Modelling
- **TF-IDF**: Term Frequency - Inverse Document Frequency
- **US**: United States
- **UTC**: Coordinated Universal Time (Temps Universel Coordonné)
- **VADER**: Valence Aware Dictionary and sEntiment Reasoner

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